AI Prompt-Chaining Assessment

Category : Healthcare
Scenario : Transforming Doctor's Notes into a Clear Patient Information Sheet
Name : Chellam.M
Reg no : 99230041152
Year : 4th year

Scenario Overview

Goal: Transform rapid, medical jargon-heavy clinical progress notes into an empathetic, highly structured, 6th-to-8th-grade reading level Patient Information Sheet (Discharge / Care Plan), incorporating a clinical safety and clarity critique stage before finalizing.

Input: Raw, Messy Doctor's Notes (provided to the model in Step 1)

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pt 58yo M, dx: newly diagnosed T2DM, HTN unctrl, dyslipidemia. BMI 31.2. A1c
9.2%, BP 148/92, LDL 154. c/o fatigue, polydipsia, nocturia x3wks. Denies
CP/SOB/diaphoresis. Plan: start Metformin 500mg PO BID with meals (titrate to
1000mg BID after 2wks if GI tolerated; warn lactic acidosis risk, GI upset,
take w/ food). Start Lisinopril 10mg PO daily (monitor BP at home, warn dry
hacking cough, angioedema risk, check K+ / Cr in 4wks). Atorvastatin 20mg PO
QHS (report unexplained myalgias, take at night). SMBG fasting AM and 2hr
postprandial goal: fasting 80-130 mg/dL, postprandial <180. Dietary consult:
low carb / Mediterranean diet, reduce sodium <2g/day, eliminate sugary
beverages. Exercise: 150 min/wk moderate aerobic (e.g. brisk walking 30 min x 5
days). Hypoglycemia edu given: rule of 15 (15g fast acting glucose if BG <70,
recheck 15 min). Red flags / ER: severe dizziness, facial swelling/lip edema,
chest pain, SOB, BG >350 or persistent vomiting. RTC 4 weeks for repeat BMP,
A1c in 3 mos. Optho consult dilated eye exam, Podiatry diabetic foot check.
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Step 1 — Structuring & Planning (Extraction)

Goal: Establish the AI's clinical communicator persona, ingest the raw physician progress notes, identify the 5 foundational patient care domains, and outline a patient-friendly action plan while flagging high-risk medical terms and confusion points before drafting.

Prompt 1

Act as an expert clinical communication specialist and primary care physician. I am going to provide you with raw, cryptic progress notes from an outpatient encounter with a newly diagnosed Type 2 Diabetes and Hypertension patient. Your task: identify the 5 core healthcare domains the patient and their family caregiver must understand, translate all cryptic acronyms and abbreviations into plain language, and construct a structured outline for an empathetic Patient Information and Care Sheet. Flag critical clinical confusion points (e.g., distinguishing low blood sugar symptoms from high blood sugar, common vs. emergency medication side effects), and designate where structured tables (medication schedule, blood sugar targets) are vital for comprehension and patient safety. Keep the tone supportive, clear, and

actionable.
[Insert Raw Doctor's Notes Here — see Input section above]

AI Output (Intermediate Response 1)

- **1. Understanding Your Diagnosis & Test Results:** Translate T2DM, HTN, Dyslipidemia, elevated BMI, and high A1c (9.2%) into plain English terms without panic-inducing clinical jargon.
- **2. Your Daily Medication Schedule & Safety:** Metformin (blood sugar), Lisinopril (blood pressure), Atorvastatin (cholesterol). Flag: construct a dedicated Medication Table with exact timings, food requirements, and side effects. Distinguish expected mild GI upset from severe adverse events.
- **3. Blood Sugar Monitoring & The 'Rule of 15':** Self-monitoring blood glucose (SMBG) targets (fasting vs. post-meal). Flag: create a Blood Sugar Target Table and an explicit callout box for Hypoglycemia (Low Blood Sugar) protocol using the Rule of 15.
- **4. Everyday Lifestyle & Preventive Care Plan:** Nutrition (Mediterranean/low-sodium), exercise plan (30 mins brisk walk), daily foot checks, and specialist referrals (Optometry dilated eye exam and Podiatry foot exam).
- **5. Red Flag Symptoms & Follow-Up Timeline:** Immediate emergency warning signs (angioedema/lip swelling from ACE-inhibitors, extreme blood sugar, chest pain) vs. routine scheduled visits (BMP blood test in 4 weeks, A1c in 3 months).
- **Recommendation:** Include two easy-to-read reference tables — Daily Medication Schedule (Section 2) and Blood Sugar Target Guide (Section 3) — plus a dedicated Callout Warning Box for low blood sugar emergencies, as medication safety is paramount.

Step 2 — Drafting (Generation)

Goal: Feed the Step 1 structured outline back to the model and generate the comprehensive Patient Information Sheet under strict plain-language constraints, tables, callout warnings, and a self-assessment quiz.

Prompt 2

Excellent. Using only the outline you just generated, write the complete Patient Information and Care Guide. Constraints: (1) use the 5 clinical domains as major section headings; (2) under each heading use concise bullet points, each bullet strictly no more than 2 sentences, written at a 6th-to-8th grade reading level; (3) include a structured Daily Medication Table (Medication name, Purpose, How & When to Take, Key Watch-outs) and a Blood Sugar Reference Table (Measurement Timing, Target Range, What It Means); (4) add a highlighted "Critical Safety Alert: Recognizing & Treating Low Blood Sugar (Rule of 15)" callout box immediately following the monitoring section; (5) end with a 6-question "Patient Understanding Self-Quiz" (mix of situational and factual recall) with an Answer & Explanation Key; (6) ensure all medical acronyms (A1c, BP, LDL, BID, PO, QHS, BMP) are translated into clear patient instructions; (7) do not exceed 750 words for the main guide body, excluding the quiz.

AI Output (Intermediate Response 2) — Patient Care Guide Draft

1. Understanding Your Diagnosis & Lab Results

- You have been diagnosed with Type 2 Diabetes, High Blood Pressure (Hypertension), and High Cholesterol (Dyslipidemia). These conditions are very common and manageable with daily habits and medication.
- Your A1c level is 9.2% (healthy target is under 7.0%), which means your average blood sugar has been higher than normal over the past three months.
- Your blood pressure was 148/92 mmHg and your LDL cholesterol is 154 mg/dL; lowering both protects your heart, kidneys, and blood vessels from long-term strain.

2. Your Daily Medication Schedule & Safety

- Take each medicine exactly as prescribed, and do not stop or change doses without speaking to your care team.
- Always take Metformin in the middle of a meal to protect your stomach from nausea or loose stools.
- Check your blood pressure before taking your morning pill, and let us know if you develop a dry, lingering cough.

Medication	Purpose	When & How to Take	Key Tips & Side Effects
Metformin (500 mg)	Lowers blood sugar levels	1 tablet twice daily with breakfast and dinner. Dose increases to 2 tablets twice daily in 2 weeks.	Take with food to avoid stomach cramps. Mild nausea is normal early on.
Lisinopril (10 mg)	Lowers blood pressure & shields kidneys	1 tablet once daily every morning with water.	May cause dizziness if you stand up fast. Call us if you get a dry cough or swollen lips.
Atorvastatin (20 mg)	Lowers bad cholesterol (LDL)	1 tablet once daily at bedtime.	Works best overnight. Report any unexplained, severe muscle soreness right away.

3. Checking Your Blood Sugar at Home

- Test your blood sugar with your fingerstick meter twice a day: first thing in the morning before eating, and 2 hours after your largest meal.
- Keep a written log of your numbers to bring to every clinic checkup so we can tailor your care plan.
- A reading below 70 mg/dL means your sugar is dangerously low and requires instant treatment with the Rule of 15.

Time of Test	Safe Target Range	What This Means for You
Fasting (First thing in the morning before eating)	80 to 130 mg/dL	Shows how well your body manages sugar overnight while resting.
Post-Meal (2 hours after the start of a meal)	Under 180 mg/dL	Shows how your body handles carbohydrates from your food.

📌 Critical Safety Alert: Recognizing & Treating Low Blood Sugar (The Rule of 15)

Symptoms of Low Blood Sugar (<70 mg/dL): Shakiness, sweating, rapid heartbeat, dizziness, hunger, sudden confusion.

Action (The Rule of 15): Immediately eat or drink 15 grams of fast-acting sugar (e.g., 4 oz fruit juice, half a can of regular soda, or 3-4 glucose tablets).

Recheck & Repeat: Wait 15 minutes and re-test your blood sugar. If still under 70 mg/dL, take another 15 grams of sugar and retest in 15 minutes.

Once Normal: Eat a small snack containing protein and complex carbs (like peanut butter with crackers) if your next meal is more than an hour away.

4. Everyday Lifestyle & Preventive Care Plan

- Focus on a Mediterranean-style plate filled with leafy vegetables, lean proteins, beans, and whole grains while cutting back on refined white breads, pastries, and sweet drinks.
- Keep daily salt intake below 2,000 mg (about one teaspoon) to assist your blood pressure medication in keeping your heart healthy.
- Aim for 150 minutes of moderate activity each week, such as a brisk 30-minute walk 5 days a week.
- Inspect the bottoms and sides of both feet daily for cuts, blisters, or redness; schedule your comprehensive foot exam with Podiatry and your dilated eye exam with Optometry.

5. Red Flag Symptoms & Follow-Up Care

- Seek immediate emergency medical attention (Call 911) if you experience chest pain, sudden shortness of breath, facial or lip swelling, or blood sugar above 350 mg/dL with persistent vomiting.
- Call our clinic within 24 hours if your blood pressure regularly tops 160/100 mmHg, or if you feel persistent lightheadedness when getting out of bed.
- Return to our clinic in 4 weeks for a basic blood chemistry test (BMP) to confirm your kidneys and potassium levels are safe.
- We will repeat your A1c blood test in 3 months to evaluate your progress and celebrate your improvements.

Patient Understanding Self-Quiz

Test your knowledge to ensure you and your family feel confident managing your health at home:

Q1. Why must Metformin always be taken during a meal rather than on an empty stomach?

Q2. What should your blood sugar reading be first thing in the morning before eating breakfast?

Q3. If you feel shaky and your fingerstick blood sugar reads 58 mg/dL, what exact steps should you follow according to the 'Rule of 15'?

Q4. Which daily medication works best when taken at bedtime to manage cholesterol?

Q5. Name two warning symptoms that require immediate emergency medical care (Call 911) rather than waiting for a regular appointment.

Q6. Why are you being referred to an eye doctor (Optometry) and a foot doctor (Podiatry) even if you do not currently feel pain in your eyes or feet?

Answer Key & Patient Explanations

A1. Taking Metformin with food helps prevent stomach upset, nausea, and loose stools while your digestive system adapts to the medicine.

A2. Your target fasting blood sugar in the morning should be between 80 and 130 mg/dL.

A3. Consume 15 grams of fast-acting sugar (such as 1/2 cup fruit juice or 3-4 glucose tablets), rest for 15 minutes, and test your blood sugar again. If still below 70 mg/dL, repeat the 15 grams.

A4. Atorvastatin (20 mg) is taken once daily at bedtime because your liver produces the most cholesterol during the overnight hours.

A5. Swelling of the lips/face (angioedema), chest pain, severe shortness of breath, or blood sugar over 350 mg/dL with continuous vomiting require immediate emergency care.

A6. High blood sugar can silently damage the tiny blood vessels in your retinas and reduce nerve sensation in your feet without early symptoms. Annual specialist exams catch and prevent issues before vision loss or foot ulcers occur.

Step 3 — Critique and Refinement (The Required Gate)

Goal: Force the model to clinically evaluate its own Step 2 draft against a rigorous medical communication and patient safety checklist, detect potential patient misunderstandings or sentence overruns, and output a polished, final version.

Prompt 3

Before we finalize, act as a strict Medical Communication and Patient Safety Reviewer evaluating this patient sheet against the following checklist: (1) Are all 5 clinical domains thoroughly covered in logical order? (2) Are all bullet points strictly kept to a maximum of 2 sentences and written in plain language (free of unclarified medical abbreviations)? (3) Do both reference tables appear accurately (Medication Schedule with dosage titration, and Blood Sugar Targets)? (4) Does the "Critical Safety Alert" callout box clearly explain the symptoms and exact stepwise Rule of 15 protocol for hypoglycemia? (5) Are the 6 self-quiz questions directly aligned with high-risk clinical tasks (medication timing, recognizing low blood sugar, emergency red flags, preventive screenings)? First, provide your detailed critique noting where the draft passed and where it required refinement. Second, output the final, corrected excerpt of the patient guide incorporating your clinical improvements.

AI Output (Intermediate Response 3) — Clinical Safety Critique

- **Item 1 — Clinical Domain Coverage:** Pass. All 5 core domains are present in logical sequence, guiding the patient from disease understanding to daily execution, monitoring, lifestyle, and red flags.
- **Item 2 — Bullet Length & Plain Language:** Partial fail. In Section 4, bullet 4 combined daily foot checks, podiatry referral, and optometry screening into a single long compound sentence that strained the 2-sentence readability constraint. In addition, Section 2's medication table needed a clearer explicit warning regarding the 2-week Metformin dose titration.
- **Item 3 — Structured Reference Tables:** Pass. Both the Medication Schedule and Blood Sugar Target tables are clinically accurate, well-formatted, and avoid confusing Latin abbreviations (BID, PO, QHS).
- **Item 4 — Hypoglycemia Safety Alert:** Pass. The Rule of 15 callout box clearly details symptoms, 15g fast carbs, 15-minute wait time, and follow-up snacks.
- **Item 5 — Patient Quiz Alignment:** Pass. The 6 quiz questions directly test actionable safety behaviors: GI upset prevention, normal ranges, hypoglycemia rescue, cholesterol timing, angioedema/emergency symptoms, and silent vascular complications.
- **Correction Applied:** Section 4, bullet 4 is restructured and split into two distinct, clear sentences distinguishing everyday home foot inspections from scheduled annual clinical specialist visits. Section 2's medication instructions are also fortified to highlight the exact titration milestone.

Final, Corrected Patient Care Guide (Section 4 Excerpt — Corrected)

4. Everyday Lifestyle & Preventive Care Plan (Final Refined Version)

- Focus on a Mediterranean-style plate filled with leafy vegetables, lean proteins, beans, and whole grains while cutting back on refined white breads, pastries, and sweet drinks.
- Keep daily salt intake below 2,000 mg (about one teaspoon) to assist your blood pressure medication in keeping your heart healthy.
- Aim for 150 minutes of moderate activity each week, such as a brisk 30-minute walk 5 days a week.
- Inspect the bottoms and sides of your feet every evening for cuts, sores, blisters, or redness. Never walk barefoot.
- We have placed referrals for your annual dilated eye exam with Optometry and a comprehensive foot evaluation with Podiatry to safeguard your vision and circulation.

All other sections (Section 1: Diagnosis & Lab Results, Section 2: Medication Table, Section 3: Monitoring & Rule of 15 Safety Alert, Section 5: Red Flags & Follow-Up, and the 6-question Patient Quiz with Answer Key) passed the safety checklist unchanged and remain as shown in Step 2 above. Combined with this refined Section 4, this completes the final patient information sheet.